

Project Background

This project aims to develop a full-body motion mirroring system for the Unitree G1 humanoid robot. The system will capture the movements of a human user through a camera and convert the detected human pose into joint commands that can be executed by the G1 robot.

Because the human body and the G1 robot have different body proportions, joint structures, and movement ranges, human movements cannot be directly copied to the robot. The system must first identify the user's full-body skeletal pose, compare it with the robot's current pose, and then calculate and adapt the required robot joint movements.

The scope of this project includes full-body motion capture and motion retargeting, including the upper body, arms, torso, hips, and legs. However, robot balance control, fall recovery, and dynamic stability management are not included within the scope of this project, as these are already handled by the robot's existing locomotion and control system. This project focuses solely on generating the correct target motion for the robot to follow, not on the robot's physical stability while executing it.

System Process

The proposed system contains two parallel input pipelines.

The first pipeline processes the human user's movement. A camera captures live image frames, which are accessed through a camera SDK or video interface. A pose-estimation component then extracts the user's full-body skeletal structure and joint coordinates from the images. This output is defined as Input 1 (In1) and represents the target pose that the robot should imitate. The system will implement a mirror-reflection mapping: when the user raises their left hand, the robot will raise its right hand, replicating the visual experience of looking into a mirror. This mapping was selected as it best suits the physiotherapy use case, where the user faces the robot directly.

The second pipeline processes the robot's current state. The system reads the G1 robot's current joint angles and positions through the G1 SDK. These joint values are converted into a robot skeletal representation. This output is defined as Input 2 (In2) and represents the robot's current pose.

Both In1 and In2 are provided to the inverse kinematics module. The inverse kinematics module calculates how the robot's joints should move from its current pose towards the target human pose. It produces a theoretical set of robot joint angles based on the difference between the two poses.

The calculated joint angles are then passed to the motion-retargeting module. This module adapts the human movement to the physical structure of the G1 robot. It accounts for differences in body proportions, available joints, degrees of freedom, and joint movement limits.

The output is a set of full-body joint commands that the G1 robot can execute.

Finally, the adapted motion commands are sent to the G1 robot through the SDK. The robot executes the movement, and its updated joint state is read again by the system. This creates a continuous motion-mirroring pipeline.

The overall system process is:

Camera → Image Frames → Human Skeleton (In1)

G1 SDK → Robot Joint State → Robot Skeleton (In2)

In1 + In2 → Inverse Kinematics → Motion Retargeting → G1 Joint Commands → Robot Execution

The final outcome of the project will be a functional prototype that allows the G1 robot to imitate the full-body movements of a human user in real time or near real time. Safety considerations, including emergency-stop mechanisms and fail-safe behaviour, will be assessed separately as part of the project's Risk Register.